Solution Requirements

Date	4 July 2026
Team ID	xxxxxx
Project Name	Human Development Index (HDI) Prediction Using Machine Learning
Maximum Marks	4 Marks

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	*Users can securely access the HDI Prediction web application to use the prediction service.*	High
2	Authorization levels	*General users can enter HDI input values and view prediction results.*	Meduim
3	External interfaces	*The system uses the **hdi_dataset.csv** dataset and the trained **model.pkl** machine learning model.*	High
4	Transactions processing	*The system accepts Life Expectancy, Mean Years of Schooling, Expected Years of Schooling, and GNI as inputs and processes them to predict the HDI score.*	High

5	Reporting	*The system displays the predicted HDI score and its category (Very High, High, Medium, or Low HDI).*	High
6	Business rules, etc.	*All input values must be valid numeric values before prediction is performed.*	High
7	Compliance to laws or regulations	*The application should protect user input and comply with basic data privacy practices.*	Medium
8	*Other*	*The system should provide a simple and user-friendly interface for easy prediction.*	Meduim

Step 2: Non-Functional Requirements (NFR)

S.No	**NFR Category**	**Requirement Description**	**Target Metric / Acceptance Criteria**
1	Performance & Speed	*The prediction should be generated quickly after the user submits the input values.*	*Response time less than **2 seconds***
2	Scalability	*The system should support multiple prediction requests without affecting performance.*	*Supports multiple users simultaneously*

3	Security & Data Privacy	*User input should be processed securely and not stored without permission.*	*Secure handling of input data*
4	Reliability & Availability	*The application should remain available whenever the Flask server is running.*	***99% availability*** *during operation*
5	Usability & Accessibility	*The web interface should be simple, responsive, and easy to use.*	*Prediction can be completed in a few simple steps*
6	*Other*	*The code should be maintainable and easy to update with new datasets or improved machine learning models.*	Easy maintenance and future enhancements